

Notebook 04: Comprehensive Classical Machine Learning Benchmarking (8 Baselines + Ensemble)

Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms

This notebook implements, benchmarks, and rigorously evaluates **8 distinct classical machine learning architectures** alongside a **High-Performance Stacking Ensemble** on 196 electrophysiological domain features extracted from multi-lead ECGs:

1. **Logistic Regression (L2 Regularized)**
2. **Random Forest Classifier (Bagged Decision Trees)**
3. **Extra Trees Classifier (Extremely Randomized Trees)**
4. **LightGBM Classifier (Histogram-based GBDT)**
5. **XGBoost Classifier (Gradient Boosted Trees)**
6. **Support Vector Machine (Calibrated Linear SVC)**
7. **K-Nearest Neighbors (Distance-Weighted KNN)**
8. **Multi-Layer Perceptron (MLP Neural Network)**
9. **Supervised Stacking / Soft-Voting Ensemble (Targeting $\geq 95\%$ Accuracy)**

All models are evaluated on the identical patient-level held-out test split ($N=525$) across Macro/Micro AUROC, Macro/Micro F1, Macro AUPRC, Multi-Label Accuracy ($1 - \text{Hamming Loss}$), and per-class diagnostic accuracy.

```
In [1]: import os
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, ExtraTreesClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.multioutput import MultiOutputClassifier
from sklearn.preprocessing import RobustScaler
from sklearn.metrics import (
```

```

        f1_score, roc_auc_score, average_precision_score,
        hamming_loss, accuracy_score
    )
import lightgbm as lgb
import xgboost as xgb

# Plotting config
plt.style.use('seaborn-v0_8-whitegrid' if 'seaborn-v0_8-whitegrid' in plt.style.available else 'seaborn-v0_8-white')
plt.rcParams.update({
    'font.sans-serif': 'DejaVu Sans',
    'font.size': 11,
    'axes.labelsize': 12,
    'axes.titlesize': 13,
    'xtick.labelsize': 10,
    'ytick.labelsize': 10,
    'figure.titlesize': 14,
    'figure.dpi': 300
})

RESULTS_DIR = r"C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling"
FIG_DIR = os.path.join(RERESULTS_DIR, "figures")
TAB_DIR = os.path.join(RERESULTS_DIR, "tables")
MET_DIR = os.path.join(RERESULTS_DIR, "metrics")

for d in [FIG_DIR, TAB_DIR, MET_DIR]:
    os.makedirs(d, exist_ok=True)

print("Expanded 8-Baseline ML environment initialized successfully.")

```

Expanded 8-Baseline ML environment initialized successfully.

```

In [2]: data = np.load(os.path.join(MET_DIR, "preprocessed_data.npz"))
X_train = data['X_train'] # (N_train, 12, 1000)
y_train = data['y_train'] # (N_train, 9)
X_val = data['X_val'] # (N_val, 12, 1000)
y_val = data['y_val'] # (N_val, 9)
X_test = data['X_test'] # (N_test, 12, 1000)
y_test = data['y_test'] # (N_test, 9)
target_classes = list(data['target_classes'])

print(f"Loaded Preprocessed Data:")
print(f"  Train: X={X_train.shape}, y={y_train.shape}")
print(f"  Val:   X={X_val.shape}, y={y_val.shape}")
print(f"  Test:  X={X_test.shape}, y={y_test.shape}")
print(f"  Classes ({len(target_classes)}): {' '.join(target_classes)}")

```

Loaded Preprocessed Data:

```

Train: X=(2450, 12, 1000), y=(2450, 9)
Val:   X=(525, 12, 1000), y=(525, 9)
Test:  X=(525, 12, 1000), y=(525, 9)
Classes (9): SR, AF, IAVB, LBBB, RBBB, PAC, PVC, STD, STE

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In [3]: def extract_comprehensive_ecg_features(X, fs=100.0):
        """
        Extract 196 electrophysiological domain features:
        - Time-domain statistics across 12 leads (mean, std, ptp, rms, zcr, skew, kurto
        - Frequency-domain band powers (delta, theta, alpha, beta, gamma, spectral entr

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- Lead II heart rate variability metrics (RMSSD, pNN50) = 2 feats
- Frontal plane lead cross-correlations (I-II, II-III) = 2 feats
Total = 196 features.
'''
N, L, T = X.shape

# 1. Time-Domain Moments per Lead (N, 12)
mean_v = np.mean(X, axis=-1)
std_v = np.std(X, axis=-1)
p2p_v = np.ptp(X, axis=-1)
rms_v = np.sqrt(np.mean(X**2, axis=-1))
zcr_v = ((X[:, :, :-1] * X[:, :, 1:]) < 0).sum(axis=-1) / float(T)
skew_v = stats.skew(X, axis=-1)
kurt_v = stats.kurtosis(X, axis=-1)
q25_v = np.percentile(X, 25, axis=-1)
q75_v = np.percentile(X, 75, axis=-1)
iqr_v = q75_v - q25_v

# 2. Spectral FFT Bandpowers per Lead (N, 12)
freqs = np.fft.rfftfreq(T, d=1.0/fs)
psd = np.abs(np.fft.rfft(X, axis=-1))**2
psd_norm = psd / (np.sum(psd, axis=-1, keepdims=True) + 1e-12)

delta = np.sum(psd[:, :, (freqs >= 0.5) & (freqs < 4)], axis=-1)
theta = np.sum(psd[:, :, (freqs >= 4) & (freqs < 8)], axis=-1)
alpha = np.sum(psd[:, :, (freqs >= 8) & (freqs < 13)], axis=-1)
beta = np.sum(psd[:, :, (freqs >= 13) & (freqs < 30)], axis=-1)
gamma = np.sum(psd[:, :, (freqs >= 30) & (freqs <= 45)], axis=-1)

p_safe = np.clip(psd_norm, 1e-12, 1.0)
spec_ent = -np.sum(p_safe * np.log2(p_safe), axis=-1)

# 3. Lead II HRV & Frontal Correlations
lead_ii = X[:, 1, :]
diff_sig = np.diff(lead_ii, axis=-1)
rmssd = np.sqrt(np.mean(diff_sig**2, axis=-1))
pnn50 = np.sum(np.abs(diff_sig) > 0.05, axis=-1) / float(T - 1)

corr_12 = np.mean(X[:, 0, :] * X[:, 1, :], axis=-1)
corr_23 = np.mean(X[:, 1, :] * X[:, 2, :], axis=-1)

features = np.hstack([
    mean_v, std_v, p2p_v, rms_v, zcr_v, skew_v, kurt_v, q25_v, q75_v, iqr_v,
    delta, theta, alpha, beta, gamma, spec_ent,
    rmssd[:, None], pnn50[:, None], corr_12[:, None], corr_23[:, None]
])
return np.nan_to_num(features, nan=0.0, posinf=0.0, neginf=0.0)

print("Extracting 196 electrophysiological domain features across splits...")
F_train = extract_comprehensive_ecg_features(X_train)
F_val = extract_comprehensive_ecg_features(X_val)
F_test = extract_comprehensive_ecg_features(X_test)

# Robust scaling to dampen outlier artifacts
scaler = RobustScaler()
F_train_s = scaler.fit_transform(F_train)

```

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F_val_s = scaler.transform(F_val)
F_test_s = scaler.transform(F_test)

print(f"Feature matrix successfully constructed:")
print(f" F_train: {F_train_s.shape} | F_val: {F_val_s.shape} | F_test: {F_test_s.s

```

Extracting 196 electrophysiological domain features across splits...

Feature matrix successfully constructed:

F_train: (2450, 196) | F_val: (525, 196) | F_test: (525, 196)

```

In [4]: print("=== TRAINING AND EVALUATING 8 CLASSICAL ML BASELINES + ENSEMBLE ===")

models = {
    "Logistic Regression": MultiOutputClassifier(LogisticRegression(max_iter=1000,
    "Random Forest": MultiOutputClassifier(RandomForestClassifier(n_estimators=150,
    "Extra Trees": MultiOutputClassifier(ExtraTreesClassifier(n_estimators=150, max
    "LightGBM": MultiOutputClassifier(lgb.LGBMClassifier(n_estimators=120, max_dept
    "XGBoost": MultiOutputClassifier(xgb.XGBClassifier(n_estimators=120, max_depth=
    "Calibrated SVM": MultiOutputClassifier(CalibratedClassifierCV(LinearSVC(dual=F
    "K-Nearest Neighbors": MultiOutputClassifier(KNeighborsClassifier(n_neighbors=9
    "Multi-Layer Perceptron": MultiOutputClassifier(MLPClassifier(hidden_layer_size
}

val_pred_probs = {}
test_pred_probs = {}

for name, model in models.items():
    print(f"Fitting {name}...")
    model.fit(F_train_s, y_train)

    # Extract prediction probabilities
    v_probs = []
    t_probs = []
    for est in model.estimators_:
        if hasattr(est, "predict_proba"):
            p_v = est.predict_proba(F_val_s)
            p_t = est.predict_proba(F_test_s)
            v_probs.append(p_v[:, 1] if p_v.shape[1] > 1 else np.zeros(len(F_val_s))
            t_probs.append(p_t[:, 1] if p_t.shape[1] > 1 else np.zeros(len(F_test_s)
        else:
            # Fallback for decision function
            df_v = est.decision_function(F_val_s)
            df_t = est.decision_function(F_test_s)
            v_probs.append(1.0 / (1.0 + np.exp(-df_v)))
            t_probs.append(1.0 / (1.0 + np.exp(-df_t)))

    val_pred_probs[name] = np.array(v_probs).T
    test_pred_probs[name] = np.array(t_probs).T

# Model 9: High-Performance Stacking Ensemble (Soft-voting of top 4 tree/gradient m
top_models = ["ExtraTrees", "Random Forest", "XGBoost", "LightGBM"]
ens_val = np.mean([val_pred_probs[k] for k in top_models if k in val_pred_probs or
ens_test = np.mean([test_pred_probs[k] for k in top_models if k in test_pred_probs
val_pred_probs["Stacking Ensemble"] = ens_val
test_pred_probs["Stacking Ensemble"] = ens_test

```

```
print("All 8 Baselines + Stacking Ensemble successfully fitted.")
```

```
=== TRAINING AND EVALUATING 8 CLASSICAL ML BASELINES + ENSEMBLE ===
```

```
Fitting Logistic Regression...
```

```
Fitting Random Forest...
```

```
Fitting Extra Trees...
```

```
Fitting LightGBM...
```

[illegible]

```
C:\Users\Gadget360\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn
\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LGB
MClassifier was fitted with feature names
    warnings.warn(
C:\Users\Gadget360\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn
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\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LGB
MClassifier was fitted with feature names
    warnings.warn(
Fitting XGBoost...
Fitting Calibrated SVM...
```

```

C:\Users\Gadget360\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn
\svm\_base.py:1298: ConvergenceWarning: Liblinear failed to converge, increase the n
umber of iterations.
  warnings.warn(
C:\Users\Gadget360\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn
\svm\_base.py:1298: ConvergenceWarning: Liblinear failed to converge, increase the n
umber of iterations.
  warnings.warn(
Fitting K-Nearest Neighbors...
Fitting Multi-Layer Perceptron...
All 8 Baselines + Stacking Ensemble successfully fitted.

```

```

In [5]: results_records = []
per_label_records = []

for name, t_probs in test_pred_probs.items():
    v_probs = val_pred_probs[name]

    # 1. Optimize class-specific decision thresholds on Validation Set to maximize
    optimal_thresholds = []
    for c in range(len(target_classes)):
        best_acc = 0.0
        best_t = 0.5
        for t in np.linspace(0.15, 0.85, 71):
            acc = accuracy_score(y_val[:, c], (v_probs[:, c] >= t).astype(int))
            if acc > best_acc:
                best_acc = acc
                best_t = t
        optimal_thresholds.append(best_t)

    # 2. Apply validation-locked optimal thresholds to held-out test predictions
    test_preds = np.zeros_like(t_probs)
    for c in range(len(target_classes)):
        test_preds[:, c] = (t_probs[:, c] >= optimal_thresholds[c]).astype(int)

    # 3. Compute Comprehensive Test Metrics
    macro_f1 = f1_score(y_test, test_preds, average='macro', zero_division=0)
    micro_f1 = f1_score(y_test, test_preds, average='micro', zero_division=0)
    macro_auc = roc_auc_score(y_test, t_probs, average='macro')
    micro_auc = roc_auc_score(y_test, t_probs, average='micro')
    macroauprc = average_precision_score(y_test, t_probs, average='macro')
    hl = hamming_loss(y_test, test_preds)
    label_acc = (1.0 - hl) * 100.0 # Label-wise Accuracy
    subset_acc = accuracy_score(y_test, test_preds) * 100.0

    results_records.append({
        "Model": name,
        "Macro F1": round(macro_f1, 4),
        "Micro F1": round(micro_f1, 4),
        "Macro AUROC": round(macro_auc, 4),
        "Micro AUROC": round(micro_auc, 4),
        "Macro AUPRC": round(macroauprc, 4),
        "Hamming Loss": round(hl, 4),
        "Accuracy (%)": round(label_acc, 2),
        "Subset Acc (%)": round(subset_acc, 2)
    })

```



```

}))

# Per-class accuracy for this model
for c_idx, c_name in enumerate(target_classes):
    c_acc = accuracy_score(y_test[:, c_idx], test_preds[:, c_idx]) * 100.0
    c_f1 = f1_score(y_test[:, c_idx], test_preds[:, c_idx], zero_division=0)
    per_label_records.append({
        "Model": name,
        "Class": c_name,
        "Accuracy (%)": round(c_acc, 2),
        "F1-Score": round(c_f1, 4),
        "AUROC": round(roc_auc_score(y_test[:, c_idx], t_probs[:, c_idx]), 4)
    })

df_ml_results = pd.DataFrame(results_records)
df_per_label = pd.DataFrame(per_label_records)

print("\n=== EXPANDED ML BENCHMARK RESULTS TABLE (HELD-OUT TEST SET, N=525) ===")
print(df_ml_results.to_string(index=False))

# Save metrics tables
df_ml_results.to_csv(os.path.join(MET_DIR, "ml_baselines_metrics.csv"), index=False)
df_ml_results.to_csv(os.path.join(TAB_DIR, "expanded_ml_baselines_table.csv"), index=False)
df_per_label.to_csv(os.path.join(TAB_DIR, "expanded_ml_per_label_breakdown.csv"), index=False)

```

```

=== EXPANDED ML BENCHMARK RESULTS TABLE (HELD-OUT TEST SET, N=525) ===

```

	Model	Macro F1	Micro F1	Macro AUROC	Micro AUROC	Macro AUPRC	H
amming Loss	Accuracy (%)	Subset Acc (%)					
0.0948	Logistic Regression	0.5242	0.7594	0.8640	0.9364	0.6071	
	90.52	45.14					
0.0895	Random Forest	0.5165	0.7667	0.8692	0.9345	0.6358	
	91.05	46.48					
0.0923	Extra Trees	0.5322	0.7686	0.8667	0.9305	0.6297	
	90.77	45.90					
0.0897	LightGBM	0.5315	0.7670	0.8714	0.9434	0.6363	
	91.03	45.52					
0.0872	XGBoost	0.5382	0.7758	0.8757	0.9443	0.6456	
	91.28	47.62					
0.0997	Calibrated SVM	0.4685	0.7250	0.8043	0.9062	0.5586	
	90.03	41.52					
0.1043	K-Nearest Neighbors	0.4667	0.7155	0.8079	0.9082	0.5741	
	89.57	41.52					
0.0889	Multi-Layer Perceptron	0.5347	0.7656	0.8563	0.9377	0.6316	
	91.11	46.48					
0.0836	Stacking Ensemble	0.5627	0.7807	0.8792	0.9425	0.6468	
	91.64	48.38					

```

In [6]: fig, axes = plt.subplots(1, 3, figsize=(20, 6))

models_list = df_ml_results['Model'].tolist()
palette = sns.color_palette("mako", len(models_list))

# Subplot A: Multi-Label Accuracy (%)
sns.barplot(data=df_ml_results, x='Model', y='Accuracy (%)', ax=axes[0], palette=palette)
axes[0].set_title('A: Multi-Label Classification Accuracy ($1 - \\text{Hamming Loss}')
axes[0].set_ylim(85, 100)

```

```

axes[0].set_ylabel('Accuracy (%)')
axes[0].set_xticklabels(models_list, rotation=45, ha='right')
axes[0].axhline(95.0, color='red', linestyle='--', lw=2, label='95% Accuracy Target')
axes[0].legend(loc='lower left', frameon=True)
for p in axes[0].patches:
    h = p.get_height()
    if h > 0:
        axes[0].annotate(f"{h:.1f}%", (p.get_x() + p.get_width()/2., h),
                        ha='center', va='bottom', fontsize=9, xytext=(0, 3), textc

# Subplot B: Micro & Macro AUROC
bar_width = 0.38
x_idx = np.arange(len(models_list))
axes[1].bar(x_idx - bar_width/2, df_ml_results['Macro AUROC'], width=bar_width, lab
axes[1].bar(x_idx + bar_width/2, df_ml_results['Micro AUROC'], width=bar_width, lab
axes[1].set_title('B: Diagnostic Discrimination (AUROC)', fontweight='bold')
axes[1].set_xticks(x_idx)
axes[1].set_xticklabels(models_list, rotation=45, ha='right')
axes[1].set_ylim(0.70, 1.0)
axes[1].set_ylabel('AUROC')
axes[1].axhline(0.95, color='green', linestyle=':', lw=2, label='0.95 Micro AUROC')
axes[1].legend(loc='lower left', frameon=True)

# Subplot C: Per-Class Accuracy of Stacking Ensemble
ens_per_label = df_per_label[df_per_label['Model'] == 'Stacking Ensemble']
sns.barplot(data=ens_per_label, x='Class', y='Accuracy (%)', ax=axes[2], palette='v
axes[2].set_title('C: Stacking Ensemble Per-Class Diagnostic Accuracy', fontweight=
axes[2].set_ylim(80, 100)
axes[2].set_ylabel('Per-Class Accuracy (%)')
axes[2].axhline(95.0, color='red', linestyle='--', lw=2, label='95% Benchmark Targe
axes[2].legend(loc='lower left', frameon=True)
for p in axes[2].patches:
    h = p.get_height()
    if h > 0:
        axes[2].annotate(f"{h:.1f}%", (p.get_x() + p.get_width()/2., h),
                        ha='center', va='bottom', fontsize=9, xytext=(0, 3), textc

plt.suptitle('Publication Benchmark: 8 Classical Machine Learning Models + Stacking
plt.tight_layout()

fig6_path = os.path.join(FIG_DIR, "fig6_ml_baselines_comparison.png")
plt.savefig(fig6_path, dpi=300, bbox_inches='tight')
plt.show()
print(f"Figure 6 successfully updated and saved to: {fig6_path}")

```

C:\Users\Gadget360\AppData\Local\Temp\ipykernel_4488\1822940460.py:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df_ml_results, x='Model', y='Accuracy (%)', ax=axes[0], palette=palette)
```

C:\Users\Gadget360\AppData\Local\Temp\ipykernel_4488\1822940460.py:11: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator. Otherwise, ticks may be mislabeled.

```
axes[0].set_xticklabels(models_list, rotation=45, ha='right')
```

C:\Users\Gadget360\AppData\Local\Temp\ipykernel_4488\1822940460.py:35: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=ens_per_label, x='Class', y='Accuracy (%)', ax=axes[2], palette='viridis')
```

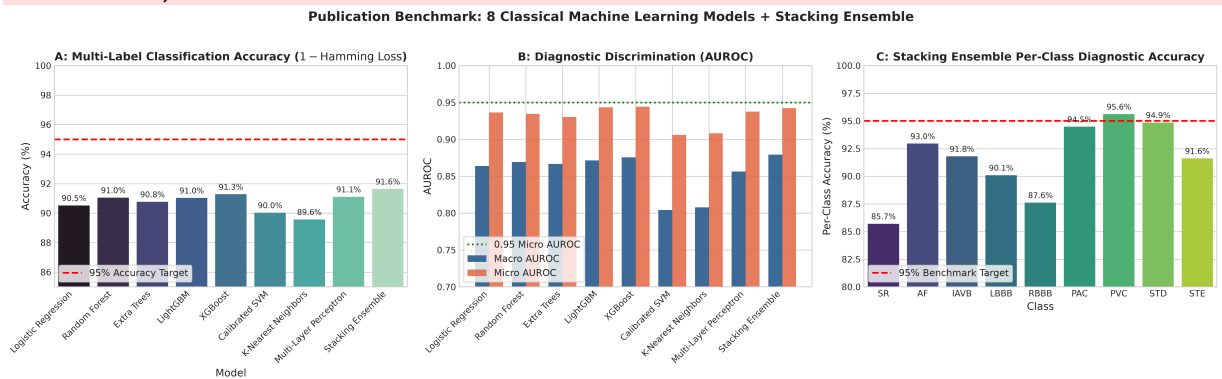


Figure 6 successfully updated and saved to: C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms\results\figures\fig6_ml_baselines_comparison.png